Prioritizing Pharmacovigilance Follow-up with Machine Learning

An Empirically Calibrated Synthetic-Data Study Using FDA FAERS

Xiao Tao Ulaş Ozan Ekmekçi Kartik Vinayak Oak

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1 Business Understanding

1.1 Business Context and Motivation

Pharmacovigilance (PV) involves the continuous monitoring and assessment of adverse drug reactions to support medication safety. Spontaneous adverse-event reports often contain incomplete information at initial submission, making follow-up important for obtaining additional details and improving case assessment. However, PV teams face practical constraints, including limited specialist capacity and difficulties obtaining responses from reporters.

Vemula et al. (2024) reported a first follow-up response rate of 32.7% among 11,980 cases in an industry PV setting, including 30.49% among reports classified as non-serious. These figures illustrate the challenge of obtaining responses; they do not measure the probability of obtaining additional serious information. FDA E2D(R1) guidance and EMA Good Pharmacovigilance Practices Module VI support appropriately prioritized, targeted follow-up and documented follow-up activities (U.S. Food and Drug Administration 2026a; European Medicines Agency 2017).

These considerations motivate using predictive analytics to help allocate limited discretionary follow-up capacity. The project investigates whether information available at initial case intake can help identify reports that are more likely to contain documented outcome information in subsequent follow-up. The model is intended to support, rather than replace, PV judgment or required follow-up activities. Related studies have explored seriousness triage, case utility, and signal prioritization (Bergman et al. 2023; Kara et al. 2023), although their prediction targets and operational settings differ from this study.

1.2 Machine Learning Objective

The central research question is: **Can information available in an initial adverse-event report predict whether outcome information will be documented in a subsequent follow-up report?**

We formulate this as a supervised binary classification task. The reference target, `new_serious_info_followup`, indicates whether at least one OUTC record is documented in a matched subsequent follow-up report (`Y = 1`) or no OUTC record is documented despite observed follow-up (`Y = 0`). Cases without observed follow-up are excluded from binary labeling.

Candidate predictors describe information available at initial intake, including patient and reporting characteristics, medication and reaction complexity, information availability, and selected clinical categories. Only baseline information is used for prediction to prevent leakage from subsequent report updates. FAERS serves as an empirical reference for constructing a synthetic case-level dataset, while the final model is trained and evaluated on synthetic data.

The model generates predicted probabilities to rank eligible cases for review. These probabilities concern the implemented outcome-documentation target and should not be interpreted as clinical deterioration risk, causal drug risk, or validated probabilities of obtaining clinically important information.

1.3 Business and Analytical Success Criteria

The primary analytical question is whether model-based ranking can identify more cases with positive outcomes under the implemented documentation-based target than random prioritization at the same review capacity. Following the identification-efficiency framework summarized in the literature review, the main operational metric is Recall@20%: the fraction of all positives found among the highest-ranked 20% of eligible cases. Under uniform random selection of K cases from N, expected recall is K/N; integer rounding makes this slightly different from nominal 20%. Precision@20% measures the positive share in the reviewed group, while Lift@20% compares that precision with overall prevalence.

ROC-AUC and PR-AUC provide complementary predictive evaluation, but do not themselves establish operational savings. A useful analytical result would show improved out-of-sample capture on held-out synthetic data with a transparent and reproducible workflow. Actual staff-time savings, faster information receipt, or clinical benefits would require prospective validation and organizational process data.

2 Data Understanding

2.1 Data Sources and Structure

The empirical reference is the FDA Adverse Event Reporting System (FAERS), a spontaneous reporting database containing adverse-event reports submitted through multiple reporting channels. Its quarterly files separate case information across related tables.

FAERS table	Main information
DEMO	Case identifiers, report versions, demographics, and reporting context
DRUG	Reported medications and drug roles
REAC	Reported adverse reactions
OUTC	Recorded patient outcome categories
INDI	Drug indications
THER	Therapy dates and treatment information
RPSR	Report sources

The `caseid` identifies an underlying case across versions; `primaryid` identifies a particular report version. Versioning makes it possible to distinguish initial-report information from subsequently documented information. Because reports are spontaneous, a recorded drug–event association does not independently establish causality.

2.2 Reference Cohort Characteristics

FAERS is used to anchor the synthetic-data design rather than as the final training dataset (U.S. Food and Drug Administration 2026b). The labeled reference cohort consists of initial reports without a matched OUTC record and with at least one observed subsequent follow-up report. The reference target identifies whether an OUTC record is documented in a matched follow-up report. The dataset contains the case counts and target distribution reported in the following table.

Table 1: Reference target distribution, recomputed from the supplied CSV.

Reference target	Cases	Percentage
Follow-up OUTC recorded	987	6.47%
No follow-up OUTC recorded	14269	93.53%
Total	15256	100.00%

The low positive prevalence creates a class-imbalanced classification problem. This supports examining precision-recall and fixed-capacity capture measures alongside ROC-AUC. The v5 reference file contains 22 candidate predictors; their construction is explained in Section 3.

2.3 Data Quality and Limitations

FAERS contains incomplete demographic, event-date, drug, and indication information. Missing values can reflect reporting practices or unavailable documentation rather than the absence of an underlying characteristic. Reports may also contain inconsistencies, changing case versions, and selective reporting.

The reference cohort is restricted to cases with observed follow-up. Cases without observed follow-up cannot reliably be assigned negative labels, introducing follow-up selection bias and limiting the representativeness of the labeled cohort. Unequal observation periods may further affect the likelihood of recording subsequent outcome information.

Previous PV triage studies illustrate a related but distinct prediction task. In particular, Lieber et al. (2023) included information that could reflect follow-up, so the reported performance does not establish accuracy using initial-intake fields alone.

The absence of an initial OUTC record does not establish that a case was clinically non-serious. Similarly, a subsequent OUTC record indicates documented outcome information rather than necessarily a newly occurring clinical event.

3 Data Preparation

3.1 Reference Cohort Construction

The reference cohort was constructed from FAERS quarterly extracts to link initial case characteristics with observed subsequent report versions.

Initial reports were identified from the 2025 Q1 DEMO table using `i_f_code = "I"`. Cases with any matched OUTC record at initial reporting were excluded using `primaryid`, resulting in a baseline cohort without recorded outcome information.

Subsequent follow-up reports were identified using `i_f_code = "F"` in quarterly DEMO extracts from 2025 Q2 through 2026 Q1 and matched to baseline cases by `caseid`. The implementation uses the follow-up indicator and case linkage without explicitly enforcing an increase in `caseversion`.

Only cases with observed follow-up were included in the labeled reference cohort. Cases without observed follow-up were excluded rather than classified as negative.

The resulting reference dataset contains one observation per eligible case. Predictors were constructed from initial-report information, while subsequent reports were used for target labeling. The observation window is defined by the available quarterly extracts rather than an identical follow-up duration for every case.

3.2 Target Definition

The binary target variable, `new_serious_info_followup`, was constructed using OUTC records linked to observed subsequent follow-up reports.

The target was defined as follows:

- **Y = 1:** At least one OUTC record is documented in a matched subsequent follow-up report.
- **Y = 0:** An observed follow-up report exists, but no matched OUTC record is documented.
- **Unlabeled:** No subsequent follow-up report is observed within the study window.

3.3 Predictor Selection and Engineering

Candidate predictors were reviewed for availability at initial intake, clinical and operational relevance, data quality, redundancy, and interpretability. The validated v5 reference dataset retains 22 candidate predictors, grouped below. Selection reflects documented construction and intended analytical use, not proven predictive importance.

Group	Candidate predictors	Count
Patient and reporting characteristics	`age_years`, `reporter_type`, `patient_sex`	3
Medication and reaction complexity	`num_drugs`, `num_reactions`, `has_concomitant_drug`	3
Information availability	`age_known`, `event_date_known`, `suspect_dose_known`, `suspect_indication_known`, `suspect_start_date_known`, `indication_known`	6
Reaction categories	`reaction_medication_error`, `reaction_systemic_symptoms`, `reaction_respiratory`, `reaction_dermatological`, `reaction_gastrointestinal`, `reaction_treatment_ineffectiveness`	6
Indication categories	`indication_oncology`, `indication_respiratory`, `indication_autoimmune`, `indication_neurological`	4
Total		**22**

Demographic and reporting fields came from DEMO; drug complexity and medication roles from DRUG; reactions and reaction counts from REAC; and indication categories from valid Primary Suspect drug indications linked through DRUG and INDI. Reporting-completeness flags indicate whether particular baseline information is available, rather than whether the clinical characteristic is absent.

The implementation requires careful interpretation of these flags. `num_drugs` counts drug records, not necessarily unique active substances; `num_reactions` counts distinct Preferred Terms. `suspect_indication_known` requires an INDI record for every Primary Suspect drug, whereas `indication_known` requires at least one valid non-placeholder indication. Dose availability requires nonmissing Primary Suspect dose text; start-date availability can include partial dates. `age_known` checks raw age availability; age conversion uses the recorded unit and can fail for unsupported units. Its agreement with converted-age missingness is checked for these saved datasets, rather than guaranteed for future inputs. Full definitions and mapping counts appear in Supplementary Data and Validation Diagnostics.

Reaction categories were built from predefined Preferred Term mappings and aggregated to case-level indicators. The extended v5 dataset includes dermatological, gastrointestinal, and treatment-ineffectiveness reaction categories, plus a neurological indication category with a documented restricted terminology mapping. Indication indicators distinguish 1 (valid matching indication), 0 (valid indication available but no category match), and NA (insufficient valid indication information). This distinction must survive downstream processing.

Potential alternatives were excluded or deferred because of missingness, high cardinality, redundancy, uncertain timing, or interpretability concerns. The identifiers `primaryid` and `caseid` were retained for traceability but are not candidate predictors. This decision-time restriction also addresses a limitation noted when comparing prior PV triage studies: models incorporating later report content cannot be assumed to perform equally well using only intake-time information (Lieber et al. 2023).

3.4 Reference Data Validation

The final reference file, `faers_reference_data_extended_v5.csv`, contains 15,256 cases and 25 fields: 22 candidate predictors, two administrative identifiers, and one binary target. The supplied handoff records a check of the saved CSV against its in-memory source. Its feature dictionary documents field roles, source tables, definitions, data types, and missing-value conventions; this render cross-checks its recorded missing counts against the CSV.

The v5 audit found that age_years was missing for 7,083 cases (46.4%), while the four indication-category indicators each had 4,356 missing values (28.6%). The latter denote insufficient valid Primary Suspect indication information and must not be recoded as zero. Nominal categories such as reporter_type and patient_sex retain their category meanings, including explicit "Unknown" values.

The supplied reference handoff reports exact computational agreement when six historical clinical indicators were independently reconstructed from the FAERS source records and recovered terminology mappings. The current render checks the saved dataset, its dictionary and its missingness conventions; it does not repeat that historical raw-file reconstruction. The six indicator counts below are recalculated from the submitted reference CSV.

Table 2: Saved reference indicator counts; historical reconstruction agreement is upstream evidence.

Indicator	Positive cases
Medication-error reactions	3458
Systemic-symptom reactions	1104
Respiratory reactions	1224
Oncology indications	1670
Respiratory indications	1401
Autoimmune/inflammatory indications	3915

The three reconstructed reaction indicators were complete, while the three reconstructed indication indicators retained the expected structural missing observations in each column. These checks support computational reproducibility of the reviewed mappings and preservation of missingness semantics. They do not establish independent clinical validity of the terminology choices or predictive benefit. The v5 reference dataset was handed off as an empirical basis for synthetic generation. Sections 3.5-3.6 extend that handoff with the supplied generation and validation workflow.

3.5 Synthetic Data Generation

3.5.a Design Objective and Principles

The synthetic dataset was constructed to represent selected characteristics of the FAERS reference data without directly copying complete case records. The objective was to preserve relevant empirical distributions, dependencies, missingness patterns, and relationships between variables.

The final synthetic dataset contains 15,256 cases and 25 variables, including 22 candidate predictors, two newly generated identifiers, and one binary target. A fixed random seed of 42 was used to ensure reproducibility. Synthetic caseid and primaryid values were generated independently of the reference identifiers.

The generation process combined empirical sampling with joint sampling blocks to preserve selected relationships between predictors. FAERS reference data informed the sampling structure, while synthetic observations were generated through random sampling mechanisms. The resulting dataset was used for machine learning, with the reference data serving as the basis for empirical calibration and validation.

3.5.b Predictor Generation Architecture

Synthetic predictors were generated using three primary joint sampling blocks to preserve selected dependencies observed in the reference data.

The first block covered reporting context and information availability, including reporter_type, patient_sex, and six information-availability indicators. These variables were sampled jointly to retain their empirical combinations.

Age was generated separately. When age_known = 0, age_years remained missing; otherwise, age was sampled from the observed reference distribution. Age groups (under 45, 45–64, and 65 or older) were used for conditional sampling but were not included as additional model predictors.

The second block jointly sampled num_drugs and has_concomitant_drug to preserve their relationship. The third block jointly sampled clinical reaction and indication indicators, conditioned on indication availability

and age group. For 50% of cases, reporter type was additionally included in the sampling conditions, with broader groups used for the remaining cases and as a fallback when reporter-specific combinations were unavailable.

This approach preserves selected empirical dependencies while acknowledging that not all relationships in the original FAERS data can be reproduced exactly.

3.5.c Synthetic Target Generation

The synthetic target was generated using a stochastic mechanism informed by selected patterns in the reference data. Five components contributed to a generation score: age, medication complexity, reporter type, selected clinical indications, and selected reactions.

Age and medication complexity each contributed up to two points, while the remaining three components contributed up to one point each, producing a total score between 0 and 7.

The score was mapped to five propensity groups, each associated with a researcher-specified probability of a positive outcome.

Table 3: Researcher-specified probabilities used for synthetic target generation.

Propensity Group	Score	Target Probability
Very Low	0	1.5%
Low	1	4.0%
Moderate	2	6.5%
Elevated	3	11.0%
High	4+	13.0%

A Bernoulli draw was then performed for each synthetic case using the probability assigned to its propensity group. Consequently, a higher score increased the probability of a positive target but did not determine the outcome with certainty.

The score weights and propensity probabilities were manually specified simulation assumptions rather than statistically estimated clinical risks. The generation score, propensity group, and target probability were excluded from model predictors to prevent direct access to the synthetic target-generation mechanism.

3.6 Synthetic Data Validation

The synthetic dataset was validated against the FAERS reference data to assess structural consistency, variable distributions, missingness patterns, selected predictor dependencies, and target behavior. The validation examined dataset dimensions, identifier uniqueness, variable domains, and consistency between reference and synthetic characteristics. Additional diagnostics assessed whether the synthetic generation process preserved the intended relationships between reporting context, medication complexity, and clinical indicators.

The separate synthetic-data validation procedure passed all 38 structural checks. The Quarto report additionally performs 33 structural checks covering the reference dataset, synthetic dataset, and cross-dataset consistency. These procedures have different scopes and should not be interpreted as identical validation exercises. The positive target rate was 6.47% in the reference dataset and 6.54% in the synthetic dataset. The synthetic positive rate also increased across the predefined propensity groups, consistent with the intended stochastic target-generation mechanism.

Clinical profile sampling was conditioned on age groups and indication availability, with reporter-aware sampling applied to 50% of cases to preserve selected empirical dependencies. However, some differences between reference and synthetic relationships remained. These findings support the structural consistency of the synthetic dataset but do not establish complete distributional equivalence or real-world predictive validity.

Table 4: Real reference and synthetic analysis datasets (computed from submitted CSVs).

Dataset	Cases	Predictors	Positives	Positive rate	Age missing	Indication unknown
FAERS reference v5	15256	22	987	6.47%	46.43%	28.55%
Synthetic v2	15256	22	997	6.54%	45.63%	28.74%

Table 5: Structural checks performed by this report.

dataset	Checks	Passed	Failed
Between datasets	5	5	0
Reference	14	14	0
Synthetic	14	14	0

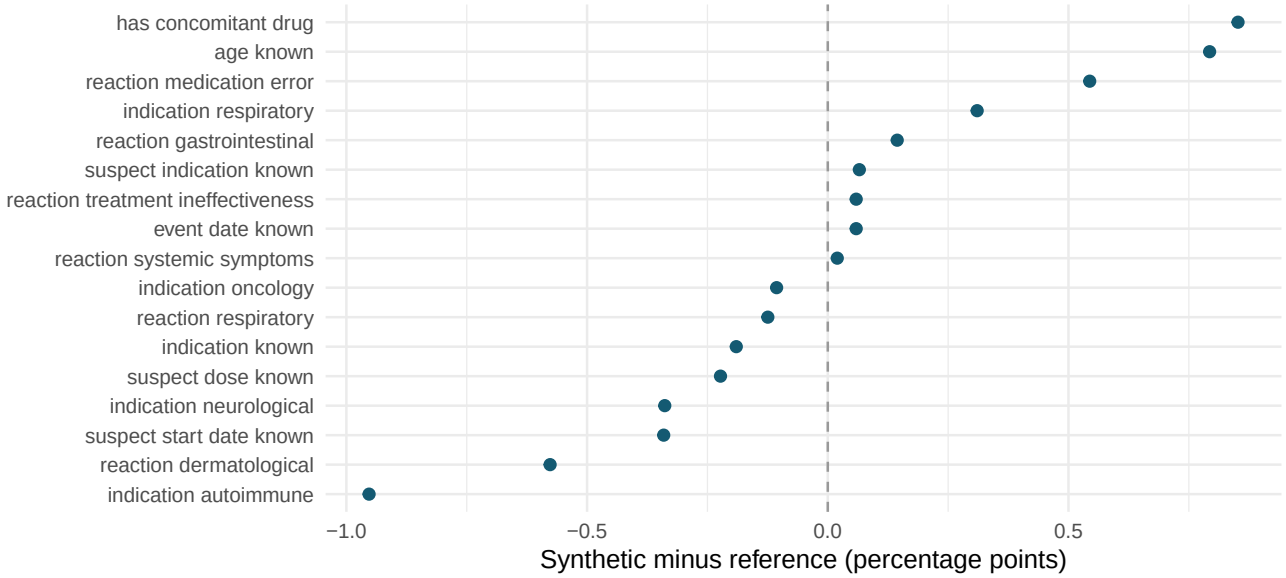


Figure 1: Binary predictor prevalence differences. Indication rates use known indications as denominator; other rates use all cases.

Hard structural agreement does not certify realism. Age conditioning preserves selected clinical relationships; the 50/50 blend only partially preserves reporter relationships, while medication and sex relationships with the clinical block can flatten. Target associations can differ because the five-family mechanism is imposed. Reusing empirical values and profiles does not establish privacy protection. The appendix provides all variable profiles and selected dependency differences, so these simplifications remain inspectable.

4 Modeling

4.1 Model Selection and Objective

Ordinary logistic regression was selected as the final model because the target variable is binary and the model produces predicted probabilities suitable for case prioritization. Rather than assigning cases to fixed positive or negative categories, the objective is to rank synthetic cases according to their estimated probability of a positive outcome under limited follow-up capacity.

Logistic regression also provides a transparent and interpretable baseline. Previous pharmacovigilance research has used logistic regression for case-level utility assessment (Muñoz et al. 2020), although its prediction target differs from this study.

The model uses 22 candidate predictors from the final synthetic dataset. Administrative identifiers (`caseid` and `primaryid`), follow-up information, and intermediate synthetic generation quantities are excluded from model inputs. The resulting predictions are evaluated within the synthetic scenario rather than interpreted as validated real-world clinical risks.

4.2 Train-Test Split and Preprocessing

The synthetic dataset was divided into training (80%) and testing (20%) subsets using a fixed random seed of 123. Stratification was performed using the binary target, with the pooling parameter set to 0.01 to retain the minority class as a separate stratum.

All preprocessing parameters were estimated using training data only, while the test set was reserved for final evaluation. This separation reduces the risk of information leakage during model development.

Table 6: Verified class stratification with seed 123 and pool = 0.01.

Partition	Cases	Positives	Positive rate
Full	15256	997	6.54%
Training	12204	797	6.53%
Test	3052	200	6.55%

Missing `age_years` values were imputed using the median age calculated exclusively from the training data. The `age_known` indicator was retained to preserve information about whether age was originally reported.

Categorical predictors, including `reporter_type` and `patient_sex`, were converted into dummy variables, and zero-variance predictors were removed. Structural missing values in the indication-category variables were recoded to zero for modeling, while `indication_known` was retained to distinguish unavailable indication information from a known nonmatching category.

These transformations were applied within the modeling workflow and did not modify the stored synthetic dataset. The training-set median age was 57.00 years. No oversampling or model tuning was performed using test outcomes.

4.3 Model Fitting and Probability Ranking

```
# Core operations of the executed function.
# Full source: Appendix, "Full Executed Modeling and Evaluation Function".
set.seed(123)
model_split <- initial_split(model_data, prop = .80,
  strata = new_serious_info_followup, pool = .01)
model_recipe <- recipe(new_serious_info_followup ~ ., data = train_data) |>
  update_role(caseid, primaryid, new_role = "id") |>
  step_impute_median(age_years) |>
  step_dummy(all_nominal_predictors()) |>
  step_zv(all_predictors())
model_spec <- logistic_reg() |> set_engine("glm") |> set_mode("classification")
model_fit <- workflow() |> add_recipe(model_recipe) |>
  add_model(model_spec) |> fit(data = train_data)
predict(model_fit, new_data = test_data, type = "prob")
```

The logistic regression model was fitted using the training data through a workflow combining the preprocessing recipe and model specification. For each held-out test case, the model generated a predicted probability of a positive synthetic outcome, which was used to rank cases from highest to lowest priority.

The objective was not to select a fixed classification threshold, but to assess whether probability-based ranking could concentrate more positive cases within limited review capacity.

Model coefficients describe associations within the assumed synthetic target-generation mechanism rather than causal clinical relationships. Performance was therefore evaluated primarily using ROC-AUC, PR-AUC, and capacity-based Recall, Precision, and Lift. The results are presented in the following Evaluation section.

5 Evaluation

5.1 Held-out Discrimination and Fixed-capacity Capture

ROC-AUC summarizes positive-negative ranking discrimination. Because positives are uncommon, trapezoidal PR-AUC is reported alongside it; this is the `yardstick::pr_auc` convention, rather than average precision (tidymodels 2026). Positive is explicitly the second factor level. Accuracy at an arbitrary 0.5 threshold would not answer the review-prioritization question.

```
auc <- c(ROC = yardstick::roc_auc_vec(test_predictions[[target]],
  test_predictions$.pred_1, event_level = "second"),
  PR = yardstick::pr_auc_vec(test_predictions[[target]],
  test_predictions$.pred_1, event_level = "second"))
stopifnot(isTRUE(all.equal(unname(auc),
  c(metrics_table$roc_auc, metrics_table$pr_auc))))
```

Table 7: Held-out synthetic test evaluation; PR-AUC uses trapezoidal integration.

Metric	Result
ROC-AUC	0.635
Trapezoidal PR-AUC	0.095
Test prevalence	6.55%
Recall at 20%	30.50%
Precision at 20%	9.98%
Lift at 20%	1.52x

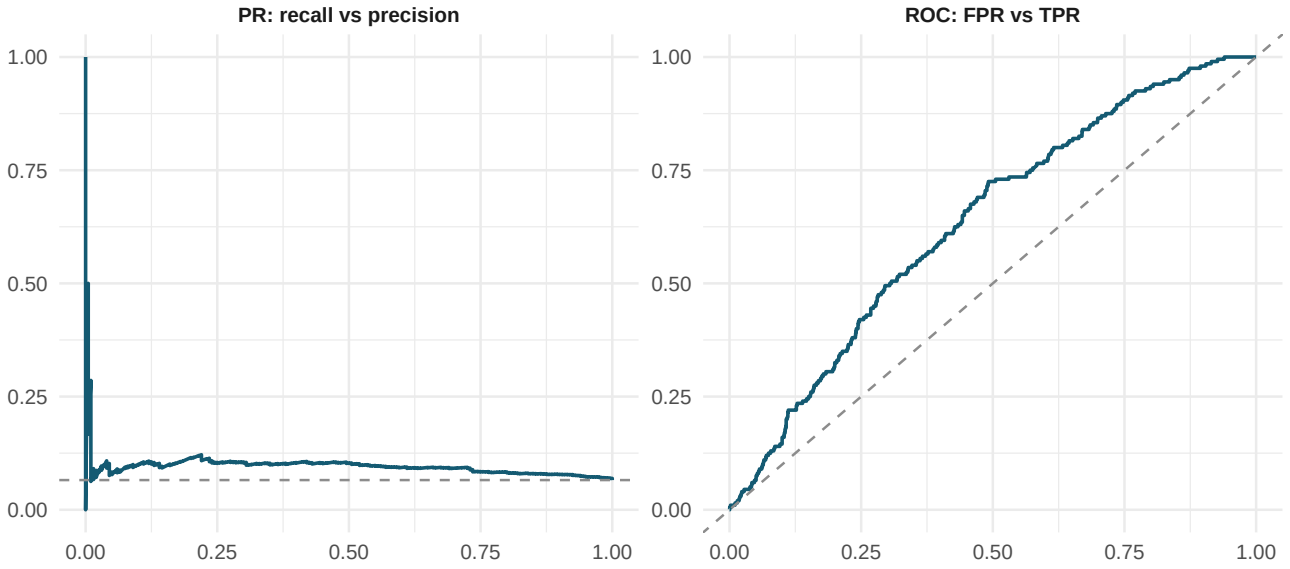


Figure 2: Synthetic test discrimination. The ROC diagonal and PR prevalence line are reference baselines.

At nominal 20% capacity, **611 / 3,052** cases are reviewed (actual capacity **20.02%**). The ranking captures **61 / 200** later-positive synthetic cases: recall **30.5%**, precision **10.0%**, and lift **1.52**. Ties are resolved deterministically by case and report identifier.

Uniform random review of the same number of cases has expected recall **20.02%** and precision equal to test prevalence. Expected captured positives are **40.0**. The Model Summary and Reproducibility section of the Appendix also provides a finite-cohort random-selection interval derived from the hypergeometric distribution. That interval describes randomness of the comparator, not confidence in model generalization.

5.2 Meaningful Extension: Capacity Sensitivity

The extension evaluates the same fixed ranking at nominal 10%, 20% and 30% review capacity. It connects the classifier to a resource-constrained decision without introducing methods beyond the course's logistic-regression scope.

Table 8: Capacity sensitivity and expected random capture at the same number of reviews.

Capacity	Reviews	Captured	Recall	Precision	Lift	Random expected
10.0%	306	29	14.5%	9.5%	1.45	20.1
20.0%	611	61	30.5%	10.0%	1.52	40.0
30.0%	916	95	47.5%	10.4%	1.58	60.0

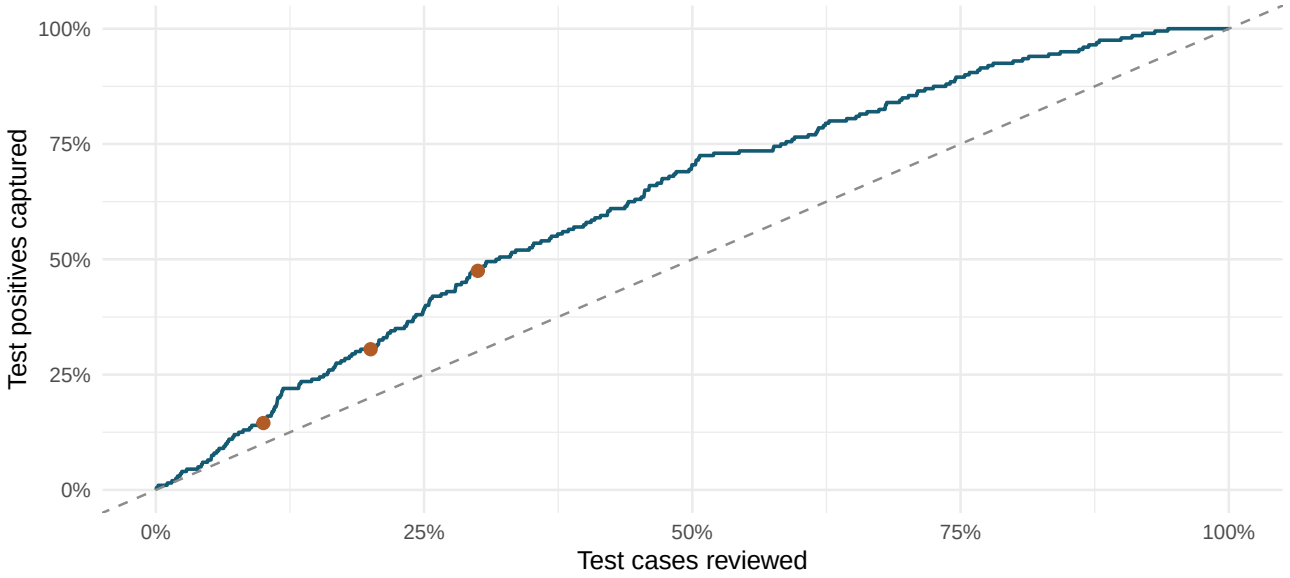


Figure 3: Positive capture against reviewed share of the synthetic test set; dashed line is expected random capture.

Increasing review capacity captures more positive synthetic cases. In the evaluated test set, precision and lift also increase across the three capacity levels, although neither metric is necessarily monotonic as capacity changes.

The analysis compares model-based and random prioritization under equal review capacity, measured by the number of cases selected. However, reviewing the same number of cases does not necessarily require the same amount of staff time.

The available literature does not provide directly transferable estimates of follow-up costs or clinically useful information yield. For example, Benaïche et al. (2025) examined the preparation of documentation requests, which represents a different intervention from case prioritization. Therefore, no monetary savings are estimated from the present results.

5.3 Limits and Business Interpretation

The model demonstrates improved identification efficiency over random ranking within the synthetic evaluation setting. The results indicate that probability-based prioritization can concentrate more positive synthetic cases within a fixed review capacity. However, these findings reflect the specified synthetic outcome-generation mechanism rather than validated performance in real-world pharmacovigilance follow-up.

The FAERS reference target represents the presence of documented OUTC information in an observed subsequent follow-up report. The synthetic target was generated using researcher-specified propensity probabilities informed by selected reference-data patterns. Consequently, model performance cannot be directly interpreted as the ability to identify newly occurring serious clinical events or to obtain clinically useful information through follow-up.

Several limitations affect generalizability. The reference cohort includes only cases with observed follow-up, introducing selection bias. Missingness patterns, terminology mappings, and unequal observation periods may also influence the reference data. Furthermore, the synthetic generator preserves selected predictor relationships but cannot reproduce all clinical and reporting complexities. Evaluation on a single held-out synthetic test set does not establish external validity.

Previous pharmacovigilance studies, including CUSP (Kara et al. 2023) and vigiRank (Caster et al. 2017), demonstrate related prioritization approaches but address different prediction targets. Before operational use, prospective validation should assess target relevance, prediction performance, and actual follow-up information yield under consistent eligibility criteria and observation windows.

6 Deployment

Deployment is proposed as a future operational workflow rather than a validated implementation. At initial case intake, mandatory follow-up requirements and professional PV judgment would first determine which cases require review. For eligible discretionary cases, a validated model could generate predicted probabilities using information available at the decision point and support priority ranking. PV specialists would retain final decision authority, including the ability to override model recommendations. A low model score would not justify disregarding mandatory follow-up requirements.

FDA E2D(R1), section 6.4, and EMA GVP Module VI, section VI.B.3, support appropriate prioritization, tailored questions and documented attempts (U.S. Food and Drug Administration 2026a; European Medicines Agency 2017). The applicable regional requirements and current organizational SOPs must be checked before use. Neither guidance validates this model or establishes a financial benefit.

Before operational implementation, the availability and consistency of the required predictors must be assessed using real organizational intake data. This is particularly important for clinical reaction and indication indicators, which depend on standardized terminology mappings and information completeness.

A prospective pilot should evaluate discrimination, calibration, and capacity-based prioritization using actual organizational cases. It should document case eligibility, review effort, follow-up attempts, response rates, and useful information obtained under consistent observation windows. Monitoring should cover missingness, predictor drift, subgroup performance, specialist overrides, and changes in reporting practices. Appropriate access controls, audit trails, version management, and manual fallback procedures would also be required.

Because the model was trained using synthetic outcomes generated through assumed propensity probabilities, its predictions cannot be interpreted as validated real-world probabilities of obtaining additional outcome information. The present study therefore demonstrates a reproducible synthetic prioritization framework rather than an operationally validated pharmacovigilance system.

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Appendix

Implementation Details and Split Sensitivity

This section documents the operational implementation and sensitivity checks supporting the reproducibility of the analysis.

The reference cohort was constructed using FAERS initial-report and follow-up indicators, case identifiers, and matched OUTC records. Initial reports with any matched OUTC record were excluded, and positive reference labels were assigned when OUTC information was documented in an observed subsequent follow-up report. The implementation does not explicitly enforce increasing case-version numbers or restrict the target to selected OUTC categories.

Synthetic predictors were generated using empirically informed sampling blocks, while target-generation scores and probabilities were researcher-specified assumptions. The main report analyzes prepared reference and synthetic datasets, with raw FAERS data preparation maintained as a separate reproducible process.

The following sensitivity analysis compares alternative train-test stratification settings. The primary model uses an 80/20 stratified split with `pool = 0.01`, while an alternative setting with `pool = 0.10` is evaluated to assess sensitivity to the pooling parameter.

Table 9: Sensitivity of model performance to alternative train-test stratification settings.

Split	Test n	Positives	ROC-AUC	PR-AUC	Recall at 20%
Stratified 80/20 split	3052	200	0.635	0.095	30.5%
Default pooling sensitivity	3052	184	0.654	0.091	34.2%

This sensitivity analysis examines whether alternative pooling settings affect the composition of the training and test sets and the resulting evaluation metrics. The primary stratification setting was specified independently of test-set performance. The comparison is therefore intended to assess methodological sensitivity rather than select the most favorable result.

Supplementary Data and Validation Diagnostics

The reference-data feature dictionary and terminology audit provide supplementary documentation of predictor construction, including variable definitions, source tables, missing-value conventions, and clinical terminology mappings. These records support methodological transparency but do not independently establish clinical validity.

The distinction between raw-age availability and converted-age values, as well as between INDI-record availability and valid indication information, is explained in Section 3.3. Definition-status fields in the feature dictionary retain items requiring further cross-checking.

Table 10: Predictor definitions (Part 1). See Section 3.3 for construction details.

Predictor	Definition
age years	Patient age expressed in years
age known	Availability of usable patient age information
event date known	Availability of event date information
reporter type	Reporter category
patient sex	Reported patient sex category
num drugs	Number of drug records associated with the report
has concomitant drug	At least one DRUG record has role_cod C
suspect dose known	Availability of suspect-drug dose information
num reactions	Number of distinct reported REAC Preferred Terms per initial report
suspect indication known	All Primary Suspect drugs have matching INDI records
suspect start date known	All Primary Suspect drugs have matched THER start-date information

Table 11: Predictor definitions (Part 2). Clinical categories use project-specific terminology mappings.

Predictor	Definition
reaction medication error	At least one REAC Preferred Term matches medication_terms
reaction systemic symptoms	At least one REAC Preferred Term matches systemic_terms
reaction respiratory	At least one REAC Preferred Term matches respiratory_terms
indication known	At least one valid non-placeholder Primary Suspect indication is recorded
indication oncology	At least one valid Primary Suspect indication matches the finalized oncology_terms mapping
indication respiratory	At least one valid Primary Suspect indication matches respiratory_indi_terms
indication autoimmune	At least one valid Primary Suspect indication matches autoimmune_indi_terms
reaction dermatological	Presence of at least one reaction term matching the category
reaction gastrointestinal	Presence of at least one reaction term matching the category
reaction treatment ineffectiveness	Presence of at least one reaction term matching the category
indication neurological	Presence of a mapped neurological Primary Suspect indication

Table 12: Clinical terminology mapping counts by indicator and source table; counts do not establish clinical validity.

variable	source_table	Term memberships
indication autoimmune	INDI + DRUG	21
indication neurological	INDI + DRUG	39
indication oncology	INDI + DRUG	97
indication respiratory	INDI + DRUG	17
reaction dermatological	REAC	46
reaction gastrointestinal	REAC	24
reaction medication error	REAC	38
reaction respiratory	REAC	15
reaction systemic symptoms	REAC	5
reaction treatment ineffectiveness	REAC	11

Table 13: All predictor summaries. Binary rates use the stated denominator; numeric summaries show median and IQR; nominal summaries show the mode.

Variable	Summary	Reference	Synthetic
age years	Median [IQR]	57.0 [35.0, 69.0]	56.0 [34.0, 69.0]
age known	1 among all	53.6%	54.4%
event date known	1 among all	33.7%	33.7%
reporter type	Most common category	Consumer 38.9%	Consumer 38.7%
patient sex	Most common category	Female 42.6%	Female 43.3%
num drugs	Median [IQR]	1.0 [1.0, 2.0]	1.0 [1.0, 2.0]
has concomitant drug	1 among all	19.3%	20.1%
suspect dose known	1 among all	81.0%	80.7%
num reactions	Median [IQR]	2.0 [1.0, 3.0]	2.0 [1.0, 3.0]
suspect indication known	1 among all	86.8%	86.9%
suspect start date known	1 among all	43.9%	43.6%
reaction medication error	1 among all	22.7%	23.2%
reaction systemic symptoms	1 among all	7.2%	7.3%
reaction respiratory	1 among all	8.0%	7.9%
reaction dermatological	1 among all	16.6%	16.0%
reaction gastrointestinal	1 among all	7.7%	7.9%
reaction treatment ineffectiveness	1 among all	6.3%	6.3%
indication known	1 among all	71.4%	71.3%
indication oncology	1 among known	15.3%	15.2%
indication respiratory	1 among known	12.9%	13.2%
indication autoimmune	1 among known	35.9%	35.0%
indication neurological	1 among known	7.5%	7.2%

Table 14: Missingness in every predictor (all-case denominator).

Variable	Reference	Synthetic
age years	46.43%	45.63%
age known	0.00%	0.00%
event date known	0.00%	0.00%
reporter type	0.00%	0.00%
patient sex	0.00%	0.00%
num drugs	0.00%	0.00%
has concomitant drug	0.00%	0.00%
suspect dose known	0.00%	0.00%
num reactions	0.00%	0.00%
suspect indication known	0.00%	0.00%
suspect start date known	0.00%	0.00%
reaction medication error	0.00%	0.00%
reaction systemic symptoms	0.00%	0.00%
reaction respiratory	0.00%	0.00%
reaction dermatological	0.00%	0.00%
reaction gastrointestinal	0.00%	0.00%
reaction treatment ineffectiveness	0.00%	0.00%
indication known	0.00%	0.00%
indication oncology	28.55%	28.74%
indication respiratory	28.55%	28.74%
indication autoimmune	28.55%	28.74%
indication neurological	28.55%	28.74%

Table 15: Selected dependence comparisons using Pearson correlation (phi for two binary variables).

Block	Pair	Reference	Synthetic	Difference
Context	age known / event date known	0.361	0.361	0.000
Context	suspect indication known / indication known	0.617	0.612	−0.005
Medication	num drugs / has concomitant drug	0.642	0.639	−0.003
Clinical	reaction systemic symptoms / reaction gastrointestinal	0.120	0.107	−0.013
Clinical	reaction dermatological / indication autoimmune	0.418	0.404	−0.014
Cross-block (simplified)	num drugs / reaction medication error	−0.046	0.012	0.057

Table 16: Outcome rates by generator-score group. Assumed probabilities are applied to synthetic cases only; reference rates are descriptive.

Dataset	Score	Cases	Positives	Observed rate	Assumed synthetic probability
Reference	0	2704	36	1.33%	1.5%
Reference	1	4449	165	3.71%	4.0%
Reference	2	3186	211	6.62%	6.5%
Reference	3	2606	281	10.78%	11.0%
Reference	4+	2311	294	12.72%	13.0%
Synthetic	0	2235	34	1.52%	1.5%
Synthetic	1	4295	174	4.05%	4.0%
Synthetic	2	4039	271	6.71%	6.5%
Synthetic	3	2775	287	10.34%	11.0%
Synthetic	4+	1912	231	12.08%	13.0%

Additional synthetic-data diagnostics are implemented in the separate validation script. Its predictive-plausibility analysis serves as a development diagnostic and is distinct from the held-out model evaluation presented in Section 5. Structural validation results should not be interpreted as evidence of clinical realism, predictive validity, or privacy protection. Its predictive-plausibility split is a development diagnostic; it does not replace Section 5 evaluation. No automatic structural PASS is treated as a certificate of clinical realism, predictive validity or privacy.

Model Summary and Reproducibility

Table 17: Selected adjusted synthetic associations; odds ratios are descriptive and reflect the generator.

Term	Log odds	Odds ratio	Std. error
age years	0.013	1.013	0.002
num drugs	0.009	1.009	0.010
reaction systemic symptoms	0.354	1.425	0.132
reaction gastrointestinal	0.031	1.031	0.136
indication oncology	0.377	1.458	0.128
indication respiratory	0.528	1.696	0.139

Table 18: Reporter-category counts and positives in the fitted split. Zero-event rare groups can yield unstable coefficients.

reporter_type	training_n	training_positives	test_n	test_positives
Consumer	4716	374	1186	92
Lawyer	48	0	6	0
Other health professional	3794	195	951	54
Pharmacist	554	31	136	11
Physician	1440	119	363	30
Unknown	1652	78	410	13

Table 19: Executed logistic model diagnostics; the Brier score is descriptive, not a deployment calibration certificate.

Check	Value
Converged	TRUE
Model matrix rank	28
Model matrix columns	28
AIC	5,721.25
Residual deviance	5,665.25
Brier score	0.06071

Table 20: R packages used by the report; full session details are exported separately.

package	version
readr	2.1.5
dplyr	1.2.1
tidyr	1.3.2
tibble	3.3.0
ggplot2	4.0.3
scales	1.4.0
rsample	1.3.2
recipes	1.4.0
parsnip	1.6.0
workflows	1.3.0
yardstick	1.4.0
broom	1.0.13
knitr	1.50

Table 21: Finite-cohort random-review capture distribution at the same number of reviews (hypergeometric quantiles).

Capacity	Model capture	Random expected	Random 2.5% quantile	Random 97.5% quantile
10.03%	29	20.1	12	28
20.02%	61	40.0	30	51
30.01%	95	60.0	48	72

Full Executed Modeling and Evaluation Function

The following R function implements the train-test split, pre-processing, logistic regression fitting, probability prediction, and evaluation procedures used in the main analysis. It is displayed for reproducibility without being executed a second time, together with the information required to reproduce the analysis.

```
# Explicit pool=0.01 retains both classes as strata; the supplied script's
# default pool=0.10 pools the approximately 6.5% positive class.
fit_and_evaluate <- function(pool_value, label) {
  model_data <- synthetic |>
    mutate(new_serious_info_followup = factor(.data[[target]], levels = c(0, 1)))
    if (pool_value == .01) stopifnot(all(prop.table(table(model_data[[target]])) >
pool_value))
  set.seed(123)
  model_split <- rsample::initial_split(model_data, prop = .80,
```

```
    strata = new_serious_info_followup, pool = pool_value)
  train_data <- rsample::training(model_split)
  test_data <- rsample::testing(model_split)
  if (pool_value == .01) {
    # For each retained stratum, rsample takes floor(0.8 * stratum size).
    stopifnot(all(as.integer(table(train_data[[target]])) ==
      floor(.8 * as.integer(table(model_data[[target]])))))
  }
  stopifnot(length(intersect(train_data$caseid, test_data$caseid)) == 0L,
    nrow(train_data) + nrow(test_data) == nrow(synthetic))
  # Categories are learned from training rows; no test summary is used.
  for (v in nominal_columns) {
    training_levels <- sort(unique(train_data[[v]]))
    if (any(!test_data[[v]] %in% training_levels)) stop("Unseen test category: ", v)
    train_data[[v]] <- factor(train_data[[v]], levels = training_levels)
    test_data[[v]] <- factor(test_data[[v]], levels = training_levels)
  }
  # Fixed semantic representation, not an estimate learned from either sample.
  train_data <- train_data |>
    mutate(across(all_of(indication_columns), ~ tidyr::replace_na(.x, 0)))
  test_data <- test_data |>
    mutate(across(all_of(indication_columns), ~ tidyr::replace_na(.x, 0)))
  model_recipe <- recipes::recipe(new_serious_info_followup ~ ., data = train_data) |>
    recipes::update_role(caseid, primaryid, new_role = "id") |>
    recipes::step_impute_median(age_years) |>
    recipes::step_dummy(recipes::all_nominal_predictors()) |>
    recipes::step_zv(recipes::all_predictors())
  model_spec <- parsnip::logistic_reg() |>
    parsnip::set_engine("glm") |> parsnip::set_mode("classification")
  model_fit <- workflows::workflow() |> workflows::add_recipe(model_recipe) |>
    workflows::add_model(model_spec) |> parsnip::fit(data = train_data)
  engine <- workflows::extract_fit_engine(model_fit)
  if (!isTRUE(engine$converged)) stop("Logistic regression did not converge.")
  test_predictions <- test_data |>
    select(all_of(c(ids, target))) |>
    bind_cols(predict(model_fit, new_data = test_data, type = "prob"))
  stopifnot(all(is.finite(test_predictions$.pred_1)),
    all(test_predictions$.pred_1 >= 0 & test_predictions$.pred_1 <= 1))
  ranked_predictions <- test_predictions |>
    arrange(desc(.pred_1), caseid, primaryid) |> mutate(rank = row_number())
  n_test <- nrow(test_predictions)
  p_test <- sum(test_predictions[[target]] == "1")
  prevalence <- p_test / n_test
  stopifnot(p_test > 0, p_test < n_test)
  metrics_table <- tibble(method = label, pool = pool_value, test_n = n_test,
    test_positives = p_test, test_prevalence = prevalence,
    roc_auc = yardstick::roc_auc(test_predictions, truth = new_serious_info_followup,
      .pred_1, event_level = "second")$.estimate,
    pr_auc = yardstick::pr_auc(test_predictions, truth = new_serious_info_followup,
      .pred_1, event_level = "second")$.estimate,
          brier_score   =   mean((as.integer(test_predictions[[target]])   ==   "1")   -
test_predictions$.pred_1)^2))
  capacity_table <- tibble(capacity = c(.10, .20, .30)) |>
    mutate(total_test_cases = n_test, reviewed_cases = ceiling(capacity * n_test),
      actual_capacity = reviewed_cases / n_test, total_true_positives = p_test) |>
    rowwise() |> mutate(positives_captured = sum(ranked_predictions[[target]][
        seq_len(reviewed_cases)] == "1"),
```

```
      recall_at_k = positives_captured / p_test,
      precision_at_k = positives_captured / reviewed_cases,
      lift_at_k = precision_at_k / prevalence,
      random_expected_positives = reviewed_cases * prevalence,
      random_expected_recall = actual_capacity,
      random_expected_precision = prevalence, random_lift = 1,
      random_capture_lower = qhyper(.025, p_test, n_test - p_test, reviewed_cases),
      random_capture_upper = qhyper(.975, p_test, n_test - p_test, reviewed_cases)) |>
    ungroup() |> mutate(method = label)
  coefficient_table <- broom::tidy(engine) |>
    mutate(odds_ratio = exp(estimate), lower_wald = exp(estimate - 1.96 * std.error),
      upper_wald = exp(estimate + 1.96 * std.error),
      unstable = !is.finite(estimate) | !is.finite(std.error) | std.error > 5)
  split_summary <- bind_rows(lapply(list(Full = model_data, Training = train_data,
     Test = test_data), function(d) tibble(rows = nrow(d), positives = sum(d[[target]]
== "1"),
      positive_rate = mean(d[[target]] == "1"))), .id = "partition") |>
    mutate(method = label)
  coefficient_diagnostics <- train_data |>
    group_by(reporter_type) |> summarise(training_n = n(),
      training_positives = sum(.data[[target]] == "1"), .groups = "drop") |>
    left_join(test_data |> group_by(reporter_type) |> summarise(test_n = n(),
     test_positives = sum(.data[[target]] == "1"), .groups = "drop"), by = "reporter_type")
   list(model_fit = model_fit, engine = engine, train_data = train_data, test_data =
test_data,
    test_predictions = test_predictions, ranked_predictions = ranked_predictions,
    metrics_table = metrics_table, capacity_table = capacity_table,
    coefficient_table = coefficient_table, split_summary = split_summary,
    coefficient_diagnostics = coefficient_diagnostics,
    training_age_median = median(train_data$age_years, na.rm = TRUE))
}
```

Open `Final_Quarto_Report.Rproj` and install the packages listed in `R/install_dependencies.R`, then run `quarto render project.qmd --to typst`. Quarto's built-in Typst engine produces a PDF without requiring a TeX installation. Paths are relative to this project. The report reads the submitted reference and synthetic CSVs, performs all analysis itself and exports fresh tables, predictions, coefficients and session information to `outputs/report/`. It does not read stored presentation metrics.

The separate `R/01_generate_synthetic_data.R` reproduces the synthetic CSV from Reference v5 using seed 42. An isolated run of that unchanged supplied script reproduced the submitted synthetic CSV byte for byte. A clean-folder render with no pre-existing model outputs also reproduced the analysis CSVs exactly. Evidence and runtime details are included under `reproducibility/`.

`R/prepare_reference.R` is preserved as supplied; its optional raw-source rebuild needs the five DEMO/OUTC quarters and initial-quarter DRUG, REAC, INDI, THER and RPSR files listed at its start, under `Data/`. Those large raw extracts are not part of the supplied report inputs. The supporting modeling script uses the `tidymodels` package, while the main Quarto analysis explicitly imports the individual component packages required for its modeling and evaluation workflow. The alternative pooling configuration is examined separately in the Implementation Details and Split Sensitivity section.

Contributions and AI Use

AI Use Disclosure

Generative AI tools were used to support code debugging and review, technical clarification, language editing, report organization, and the development and documentation of the synthetic-data generation workflow. AI assistance also supported literature checks, integration of report sections, and review of reproducibility and rendering issues.

The project authors reviewed the methodological decisions, simulation assumptions, model specifications, validation procedures, and interpretations. The authors remain responsible for the submitted analysis and its conclusions. All AI-assisted outputs must be validated against the submitted project files.

Individual Contributions Statement

Xiao Tao was the primary author of Business Understanding, Data Understanding, and reference Data Preparation preceding synthetic data generation (Sections 1, 2, and 3.1–3.4).

Ulaş Ozan Ekmekçi was primarily responsible for synthetic-data generation, synthetic-data validation, and Modeling (Sections 3.5–3.6 and 4).

Kartik Vinayak Oak was the primary author of Evaluation (Section 5), including performance metrics, capacity-based prioritization, the literature-supported evaluation framework, and interpretation of findings and limitations.

The remaining sections were developed collaboratively by the group.